

yolov8n Hyperparameter Tuning Report

Model:	yolov8n
Dataset:	bdd100k_yolo_limited
Timestamp:	2025-11-23 10:08:01
Total Trials:	3
Best Trial:	1
Best mAP@0.5:	0.0898

Optimization Configuration

Parameter	Value
Total Trials	3
Epochs per Trial	2
Batch Size	32
Image Size	640
Startup Trials	1
Device	cuda
Dataset Path	/computer_vision_yolo/bdd100k_yolo_limited
Number of Classes	10

Trial Statistics

Status	Count
Completed	3
Pruned	0
Failed	0
Total	3

Best Hyperparameters

Parameter	Value
optimizer	Adam
lr0	0.000982
lrf	0.078926
momentum	0.670413
weight_decay	0.000021
warmup_epochs	5
warmup_momentum	0.496487
warmup_bias_lr	0.068397
hsv_h	0.023323
hsv_s	0.714931
hsv_v	0.788631
degrees	9.466246
translate	0.193880
scale	0.439424
shear	8.781677
perspective	0.000513
flipud	0.575831
fliplr	0.693244
mosaic	0.758995
mixup	0.568115
copy_paste	0.134570
box	6.562520
cls	2.373073
df1	2.550120

Top 10 Trials

Trial	mAP@0.5	State
1	0.0898	COMPLETE
0	0.0861	COMPLETE
2	0.0695	COMPLETE

Optimization History

Optimization history chart not found (PNG format required).

Parameter Importance

Parameter importance chart not available or could not be generated.

Final Model Performance

Metric	Value
mAP@0.5	0.1294
mAP@0.5:0.95	0.0606
Precision	0.5832
Recall	0.0821

Generated by YOLO Hyperparameter Tuning Notebook
BDD100K Dataset - Computer Vision Project